

**CSE 332 L / EEE 336 L Computer Organization and Architecture (Section 5)****Lab Outline (CO3)**

Class Timing	Classroom
M 11:20 AM – 02:30 PM	SAC 507

Faculty:

Tanjila Farah (TnF)

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Lab Instructor:

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Course Description:

This course introduces students to the basic concepts of computers, their design, and how they work. It encompasses the definition of the machine's instruction set architecture, its use in creating a program, and its implementation in hardware. We will study modern computer principles using a typical processor and emphasize system-level issues, understanding process performance, and the use of abstraction as a tool to manage complexity. We will employ both hardware and software tools to realize a small processor of our own.

Mark Distribution (tentative):

Lab Attendance:	10%
Lab Performance:	20%
Lab Report (average):	20%
Midterm Exam:	25%
Final Exam:	25%
Project:	(Separate Percentage of theory course)

Tentative Class Schedule:

Tentative Date	Experiments
6 th October, 2025	Lab 1: Design of a 2-bit Arithmetic unit (Hardware)
13 th October, 2025	Lab 2: Design of a 4-bit Binary Up-Down counter (Hardware)
20 th October, 2025	Lab 3: Design of a 4-bit Universal Shift Register (Hardware)
27 th October, 2025	Lab 4: Design of a 2-bit Logic unit. & Design of a 4-bit Multiplier Unit (Hardware)
3 rd November, 2025	MIDTERM EXAM
10 th November, 2025	Lab 5: Design of a Register File (Software)
17 th November, 2025	Lab 6: Design of an ALU (Software)
24 th November, 2025	Lab 7: Design of a single cycle Datapath for R-type & I-type Instructions (Software)
1 st December, 2025	Lab 8: Design of a Pipeline (Software)
8 th December, 2025	FINAL EXAM